

Arizona State University
School of Electrical, Computer and Energy Engineering

EEE 465/591: Photovoltaics Energy Conversion
Practice Problem Set #8

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1. Future Balance Calculation

What is the balance in an account at the end of 10 years if \$2,500 is deposited today and the account earns 5% interest?

Given:

- $PV = \$2,500$
- $i = 0.05$ (5% interest rate)
- $N = 10$ years

Formula:

$$FV = PV(1 + i)^N$$

Calculation:

$$FV = 2500(1 + 0.05)^{10}$$

$$FV = 2500(1.62889)$$

$$FV = 4072.23$$

Answer: **\$4,072.23**

2. PV System Operating & Maintenance Cost Analysis

2(a) NPV of Operating & Maintenance (Escalating Cost)

Calculate the net present value of O&M costs starting at \$100 per year, increasing at 3% annually, with 2% inflation over 25 years.

Given:

- $A_1 = \$100$ (initial cost)
- $g = 0.03$ (escalation rate)
- $i = 0.02$ (inflation/discount rate)
- $N = 25$ years (system life)

Formula (Geometric Gradient):

$$NPV = A_1 \times [(1 - ((1+g)/(1+i))^N) / (i - g)]$$

Calculation:

$$NPV = 100 \times [(1 - (1.03/1.02)^{25}) / (0.02 - 0.03)]$$

$$NPV = 100 \times [(1 - 1.2772) / (-0.01)]$$

$$NPV = 100 \times [(-0.2772) / (-0.01)]$$

$$NPV = 100 \times 27.723$$

$$NPV = 2772.30$$

Answer: **\$2,772.30**

2(b) NPV of Fixed Contract & Comparison

Calculate the NPV of a fixed O&M contract at \$125 per year and determine if it is financially beneficial.

Given:

- $A = \$125$ (fixed annual cost)
- $i = 0.02$ (inflation/discount rate)

- N = 25 years

Formula (Uniform Series):

$$NPV = A \times [((1+i)^N - 1) / (i(1+i)^N)]$$

Calculation:

$$NPV = 125 \times [((1.02)^{25} - 1) / (0.02(1.02)^{25})]$$

$$NPV = 125 \times [(1.6406 - 1) / (0.02 \times 1.6406)]$$

$$NPV = 125 \times [0.6406 / 0.03281]$$

$$NPV = 125 \times 19.523$$

$$NPV = 2440.44$$

Comparison:

Fixed Contract NPV: \$2,440.44

Escalating Cost NPV: \$2,772.30

Since \$2,440.44 < \$2,772.30, the fixed contract has a lower net present value.

Answer: The fixed contract is beneficial (Lower NPV: \$2,440.44)

Savings: \$2,772.30 - \$2,440.44 = \$331.86

Summary:

Cost Option	NPV	Decision
Escalating O&M (\$100, 3% increase)	\$2,772.30	Higher Cost
Fixed Contract (\$125/year)	\$2,440.44	Recommended
Net Savings: \$331.86		